

tions designed in line with ergonomic requirement to suit laboratory uses," he adds.

Arun Dogra, vice president-sales, marketing and support, Agilent Technologies, talks about multipurpose lab solution (MPLS) kits. These are all-in-one solutions with certain set of equipment interconnected in one ready-to-teach package that saves lecturers' time in developing and updating courses to keep pace with the latest industry trends. Kits available today are suitable for analogue electronics, RF circuit design, digital RF communication, digital systems and digital signal processing.

Manufacturers are engaged in creating industry-grade equipment for academia. These equipment are similar to the equipment already used in the industry. "We know the budget restriction of the academic labs, so we give them what will work for them. For example, a 50GHz performance network analyser may be an overkill for a student in lab; a simple economy network analyser with 9kHz to 20GHz range would do the trick for him for now. Once the student is familiar with the features, specifications and commercial use of the equipment, he can move to the next level of that equipment. Depending on his application in future, he will know what range or specifications he should look for," shares Dogra.

Academic requirements

"We expect the manufacturer to provide the service of equipment well in time. Ruggedness, low cost, low maintenance time, easy availability of spare parts and local support and training are the things that we look for"—Prof. Khanna, Thapar University

"We expect delivery and installation of the equipment within time and demonstration/training support from the manufacturers. In most labs, equipment servicing and maintenance after installation is also desirable on suitable terms and conditions"—Dr L.K. Maheshwari, BITS, Pilani

Another new thing happening today is the concept of creating an equivalent virtual world. "Virtual instrumentation and simulation consisting of modular hardware, signal processing units and graphical programming provides a quick learning platform to the students," explains Dr Maheshwari. Lots of software for design and simulation have become a part of academic labs today.

Upcoming high-end research labs

Research and development is very important in accreditation of technical universities. Research labs are set up for postgraduate and doctorate programmes in the universities. As there is a huge amount of investment involved in research, the universities seek funding from consortium, associations, organisations and industries. The test and measurement equipment used in these labs exhibit very high precision, sensitivity and repeatability, high

bandwidths, connectivity to external source and reliability.

One of the most promising upcoming fields is the nano technology. "In nano technology, we are talking about measurements of the order of 10^9 . We require very high-precision and high-resolution equipment for these measurements," says Singh. Also, to sense such a tiny value of signal, you need nano probes and high-precision multimeters.

"Security technology, a very interesting subject, is evolving with great importance given to topics like biometrics and pattern recognition. The labs usually have a lot of different types of sensing and measurement systems," shares Bhushan. In these labs, lots of software are used for analysis and simulation.

Advanced communication labs also demand a special set of equipment. "The fastest growing field is telecommunication and the technology is changing rapidly. For example, multiple-input and multiple-output

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